

EXPERIMENT 13

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Semester: 5
Subject Name: ADBMS

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Aim - To study Amazon RDS, create a database instance on AWS RDS, understand security groups, and connect a cloud RDS database to a local database client such as pgAdmin.

Software Used:

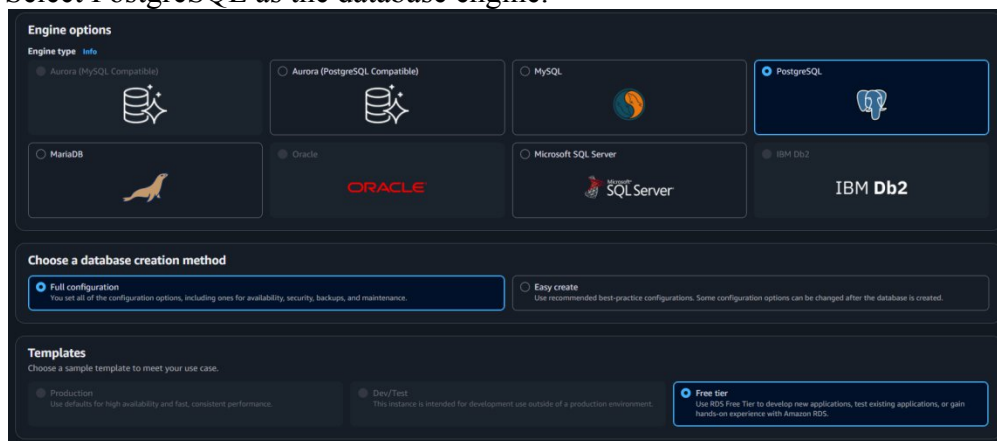
- Amazon RDS
- pgAdmin
- AWS Management Console
- Amazon EC2

Theory:

Amazon Relational Database Service (Amazon RDS) is a managed cloud service that simplifies the setup, operation, and maintenance of relational databases. It supports database engines such as PostgreSQL, MySQL, and MariaDB. An RDS database instance can be accessed remotely using its endpoint and port. AWS Security Groups act as virtual firewalls that control inbound and outbound traffic to the database. Tools such as pgAdmin can connect to an RDS PostgreSQL instance using the database endpoint, username, password, and configured security rules.

Steps:

1. Open the AWS RDS Console.
2. Select PostgreSQL as the database engine.



3. Enter the DB instance identifier.

Settings

Engine version [Info](#)
View the engine versions that support the following database features.

▼ **Hide filters**

☐ **Show only versions that support the Multi-AZ DB cluster** [Info](#)
Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.

Engine version
PostgreSQL 18.3-R2 ▼

☐ **Enable RDS Extended Support** [Info](#)
Amazon RDS Extended Support is a [paid offering](#). By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [RDS for PostgreSQL documentation](#).

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.
KRG-DB
The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.
postgres
1 to 16 alphanumeric characters. The first character must be a letter.

4. Choose self-managed credentials and set the master password.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ **Managed in AWS Secrets Manager - most secure**
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ **Self managed**
Create your own password or have RDS create a password that you manage.

☐ **Auto generate password**
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength [Average](#)
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password [Info](#)

5. Select the required instance type.

Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ **Hide filters**

☐ **Include previous generation classes**

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ **Burstable classes (includes t classes)**

Instance type
db.t4g.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps ▼

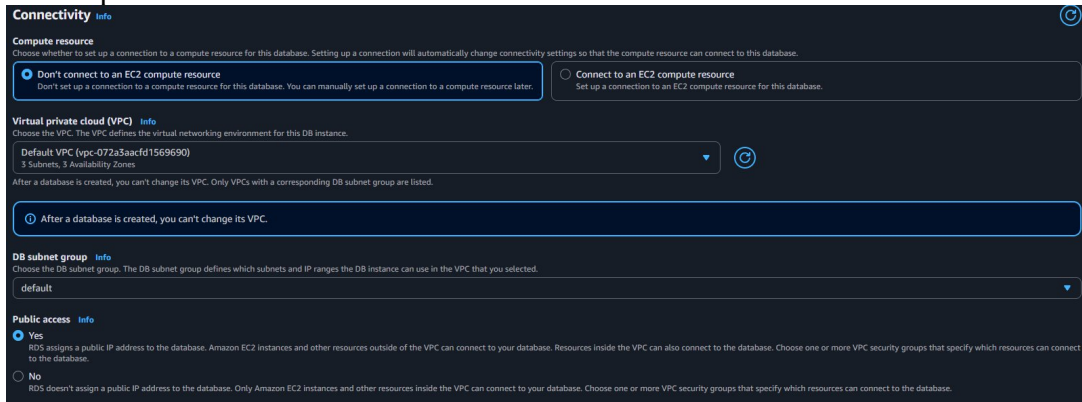
Storage

Storage type [Info](#)
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp2)
Baseline performance determined by volume size ▼

Allocated storage [Info](#)
20 GiB
Allocated storage value must be 20 GiB to 6,144 GiB

► **Additional storage configuration**

6. Enable public access.



Connectivity info

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) info
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-072a3acfd1569690)
3 Subnets, 3 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

DB subnet group info
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

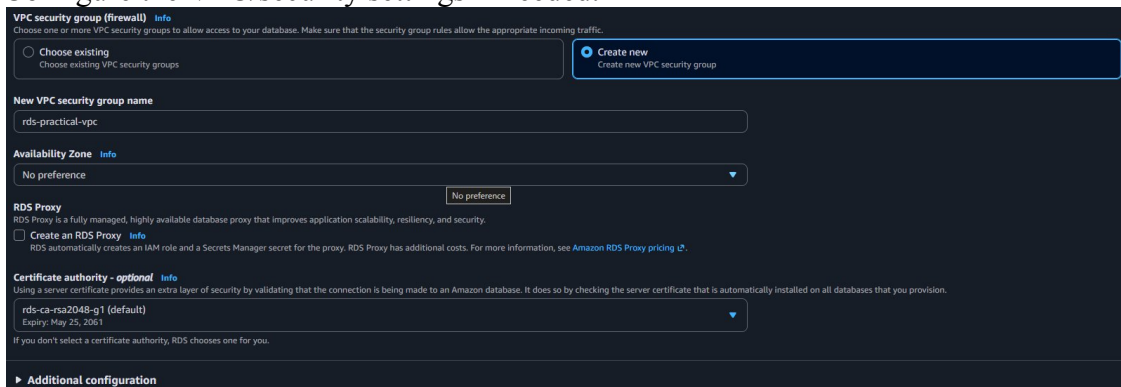
default

Public access info

☒ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

7. Configure the VPC/security settings if needed.



VPC security group (firewall) info
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☐ **Choose existing**
Choose existing VPC security groups

☒ **Create new**
Create new VPC security group

New VPC security group name
rds-practical-vpc

Availability Zone info
No preference

RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ **Create an RDS Proxy** info
RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional info
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

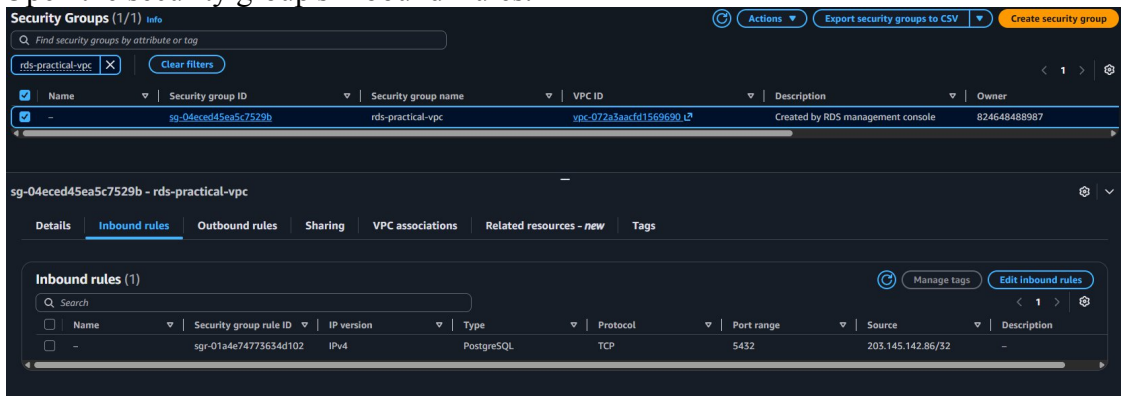
rds-ca-rsa2048-g1 (default)
Expiry: May 25, 2061

If you don't select a certificate authority, RDS chooses one for you.

Additional configuration

8. Wait until the database becomes Available.

9. Open the security group's Inbound Rules.



Security Groups (1/1) info

Find security groups by attribute or tag

rd-s-practical-vpc

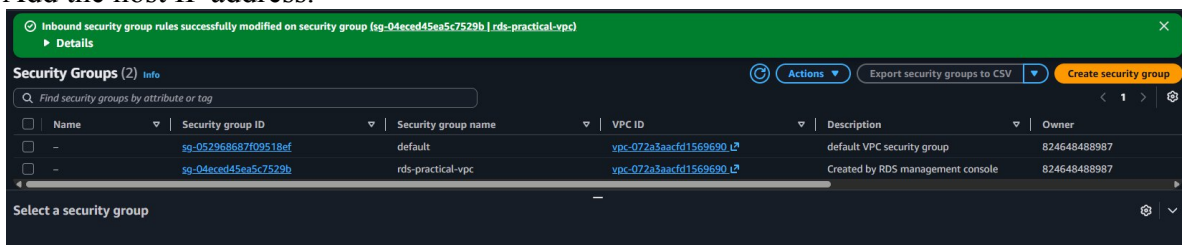
Name	Security group ID	Security group name	VPC ID	Description	Owner
-	sg-04eced45ea5c7529b	rds-practical-vpc	vpc-072a3acfd1569690	Created by RDS management console	824648488987

sg-04eced45ea5c7529b - rds-practical-vpc

Inbound rules (1)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-01a4e74773634d102	IPv4	PostgreSQL	TCP	5432	203.145.142.86/32	-

10. Add the host IP address.



Inbound security group rule successfully modified on security group (sg-04eced45ea5c7529b | rds-practical-vpc)

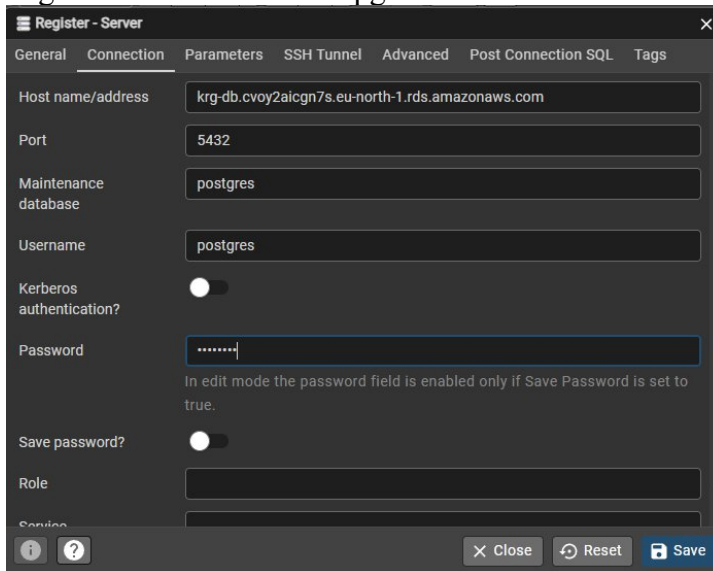
Security Groups (2) info

Find security groups by attribute or tag

Name	Security group ID	Security group name	VPC ID	Description	Owner
-	sg-052968687f09518ef	default	vpc-072a3acfd1569690	default VPC security group	824648488987
-	sg-04eced45ea5c7529b	rds-practical-vpc	vpc-072a3acfd1569690	Created by RDS management console	824648488987

Select a security group

11. Register the RDS server in pgAdmin.

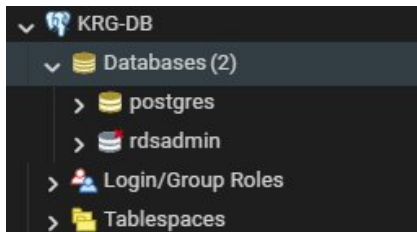


The 'Register - Server' dialog box in pgAdmin is shown with the following fields and values:

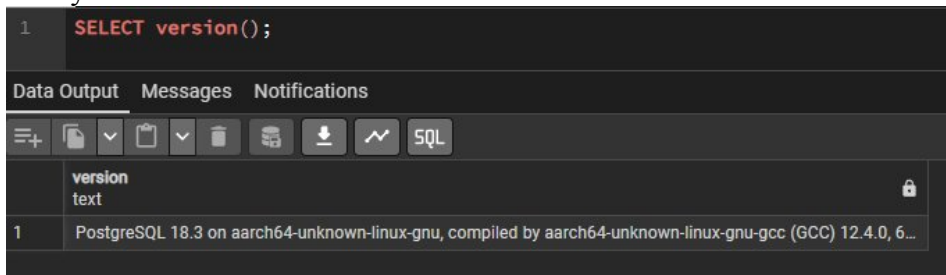
- Host name/address: krg-db.cvoy2aicgn7s.eu-north-1.rds.amazonaws.com
- Port: 5432
- Maintenance database: postgres
- Username: postgres
- Kerberos authentication?: ☐
- Password: (masked with dots)
- Save password?: ☐
- Role: (empty)
- Service: (empty)

Buttons at the bottom: Close, Reset, Save.

12. Confirm that the database is added successfully.



13. Verify the database connection.



Learning Outcomes:

1. Learnt how to create and configure an Amazon RDS PostgreSQL instance.
2. Understood how security groups and inbound rules control database access.
3. Learnt how to connect an AWS RDS database to pgAdmin.